Sets, relations, functions, graphs, and trees exercises

All questions are original. Graphs are finite, simple, and undirected unless the question specifies directions. Free trees are nonempty, and heights count edges.

B-E01 Set logic and counting

Let $U = \{0, \dots, 9\}$, $A = \{0, 2, 4, 6, 8\}$, $B = \{1, 2, 3, 4\}$, and $C = \{2, 5, 8\}$. Find $A \setminus (B \cup C)$ and prove elementwise that it equals $(A \setminus B) \cap (A \setminus C)$. Compute the union size and all intersection counts.

B-E02 Equivalence classes and partitions

On $S = \{-3, -2, \dots, 5\}$, define aRb when $4 \mid (a - b)$. Prove the equivalence axioms, list the classes, and count the ordered pairs in R. Prove generally that intersecting equivalence classes are equal; enumeration of this example is insufficient.

B-E03 Counterexamples among relation properties

On $A = \{a, b, c\}$, give a relation for each requirement:

1. Symmetric and transitive, but not reflexive.
2. Reflexive and transitive, but not symmetric.
3. Both symmetric and antisymmetric.

Why are symmetry and antisymmetry not complementary?

B-E04 Poset covers and extrema

Order $S = \{1, 2, 3, 6, 9\}$ by divisibility. Give all covers, minimal and maximal elements, and any minimum and maximum. Prove uniqueness of a maximum. What changes when 1 is removed?

B-E05 Codomain and inverse

Let $f : \mathbb{Z} \to \mathbb{Z}$ have $f(x) = 2x - 1$, and let $g : \mathbb{Z} \to \mathbb{Z}$ have $g(x) = \lfloor x/2 \rfloor$. Determine whether each function is injective or surjective, give its range, and decide whether an inverse function exists on the whole codomain. Find the set preimage $f^{-1}(\{-1, 1\})$. Distinguish an inverse relation from an inverse function.

Sets, relations, functions, graphs, and trees exercises (continued)

B-E06 Directed degrees and SCCs

Let $V = \{a, b, c, d, e\}$, with arcs $a \to b$, $b \to a$, $b \to c$, $c \to d$, $d \to c$, and $e \to e$. List indegrees, outdegrees, SCCs, components after ignoring direction, and the induced edges on $\{b, c, d\}$. Why must the loop at e be removed in a simple undirected version?

B-E07 Equal degrees do not prove isomorphism

Compare C_6 and two disjoint triangles. Give n, m, and the degree multisets, then prove nonisomorphism. State the full "if and only if" condition for isomorphism. For finite simple graphs with a vertex bijection and equal vertex and edge counts, is preserving edges in one direction sufficient? Prove your answer.

B-E08 Six conditions and boundaries

For nonempty simple undirected graphs, prove that being connected with $m = n - 1$ is equivalent to being a tree. Do not assume the tree edge count that must be proved. Give an example with $m = n - 1$ that is not a tree. Explain why $n = 0$ is excluded from the six-condition theorem.

B-E09 Tree and forest operations on one example

Let $V = \{a, b, c, d, e, f\}$ and $E = \{ab, bc, bd, de, ef\}$. Find the unique simple path from a to f, the components after deleting bd, and the cycle after adding cf. Root the tree at d. List each vertex's parent, depth, height, and child count, and compare its graph degree.

B-E10 Binary empty slots and counts

The root r has only a right child s, and s has only a left child t. List the left and right slots, height, and logical `NIL` positions. Give I, L for the extended full tree. Prove the general count of $n + 1$ missing slots, and explain why sharing a `NIL` object does not change it.

B-E11 Full versus complete k-ary

A full ternary tree has $I = 5$. Find L and the number of edges. Is this possible if all leaves must also have equal depth? Prove $L = (k - 1)I + 1$ separately from the level formulas for a complete tree.

Sets, relations, functions, graphs, and trees exercises (continued)

B-E12 Bipartite graphs and odd cycles

Prove that a finite undirected graph is bipartite if and only if it has no odd simple cycle. For a quadrilateral with one diagonal, identify the coloring conflict, and give the sides after removing the diagonal. If your proof uses an odd closed walk, prove that it contains an odd simple cycle.

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

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https://github.com/xinhi6716/computer-science-self-study

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